

A cluster of algebra, geometry and combinatorics

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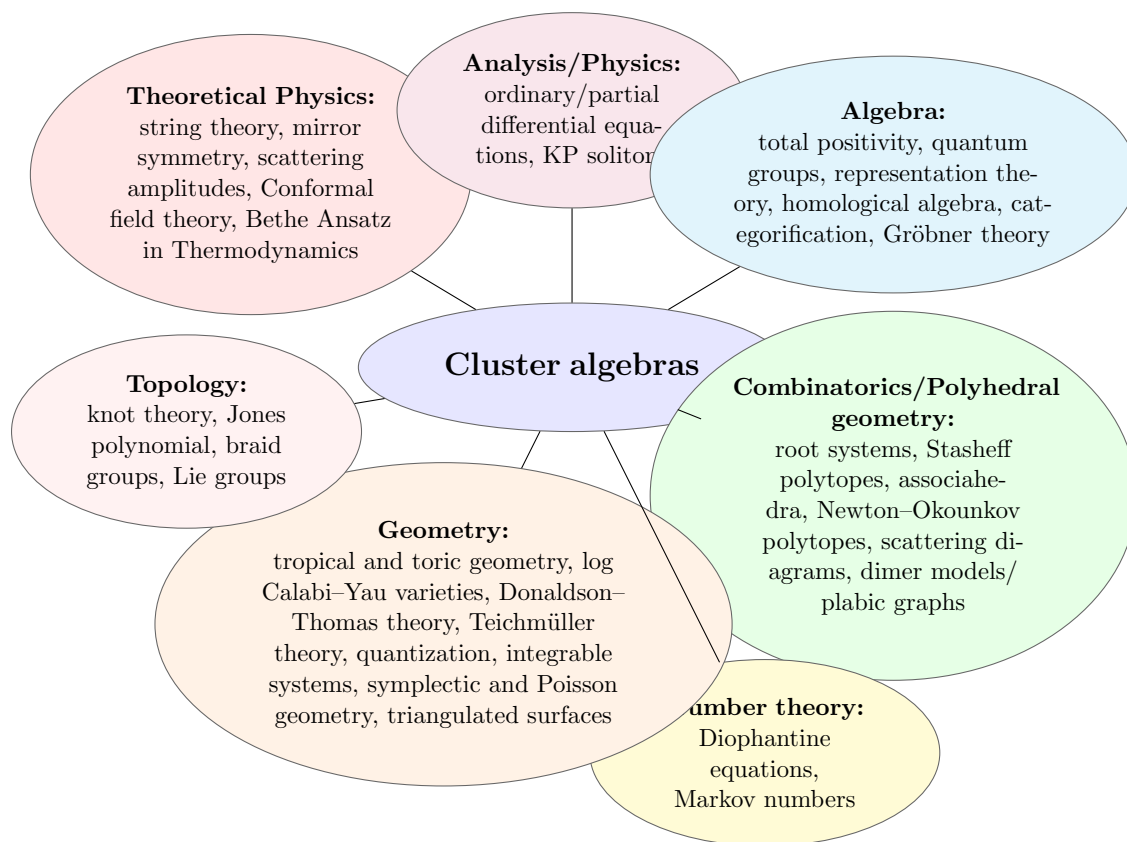
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1 Lecture 1 — Cluster Algebras



History

- Two Russian Mathematicians, Sergei Fomin and Andrei Zelevinsky, defined cluster algebras in 2001.
- They observed the *cluster structure* in the theory of total positivity of matrices, in quantum groups and their canonical bases.
- They now have their own classification 13F60.
- As of 2018 there were more than 1,755 articles and since 2003 there have been more than 139 events on cluster algebra.

1.1 The origin of cluster algebras: total positivity

Definition 1. Let $M = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \end{pmatrix} \in \mathbb{R}^{2 \times 4}$ of rank 2. It is called **totally positive** if all 2-minors of M are positive real numbers:

$$\Delta_{ij}(M) := \det \begin{pmatrix} a_i & a_j \\ b_i & b_j \end{pmatrix} = a_i b_j - a_j b_i > 0 \quad \text{for all } i < j.$$

Example 1. For example, for $M = \begin{pmatrix} 1 & 0 & -3 & -5 \\ 0 & 2 & 1 & 1 \end{pmatrix}$ we compute

$$\Delta_{12} = 2, \Delta_{13} = 1, \Delta_{14} = 1, \Delta_{23} = 6, \Delta_{24} = 10, \Delta_{34} = 2.$$

Exercise: Every $M \in \mathbb{R}^{2 \times 4}$ satisfies $\Delta_{12}\Delta_{34} + \Delta_{14}\Delta_{23} = \Delta_{13}\Delta_{24}$.

In the example: $2 \cdot 2 + 1 \cdot 6 = 1 \cdot 10$.

Question: How can we *efficiently* verify whether M is totally positive or not?

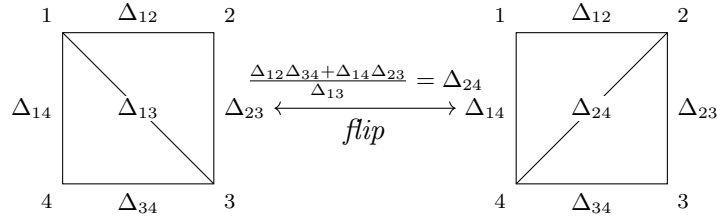
Observation: $M \in \mathbb{R}^{2 \times 4}$ is totally positive if and only if $\Delta_{12}, \Delta_{13}, \Delta_{14}, \Delta_{23}, \Delta_{34} > 0$. Indeed, in this case also

$$\Delta_{24} = \frac{\Delta_{12}\Delta_{34} + \Delta_{14}\Delta_{23}}{\Delta_{13}} > 0.$$

The set $\{\Delta_{13}, \Delta_{12}, \Delta_{14}, \Delta_{23}, \Delta_{34}\}$ is called a *positivity test*.

Question: How can we find positivity tests?

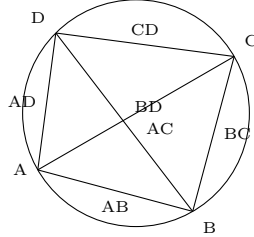
For $\mathbb{R}^{2 \times 4}$ we have another positivity test: $\{\Delta_{24}, \Delta_{12}, \Delta_{14}, \Delta_{23}, \Delta_{34}\}$. Both can be visualized using triangulations of the square:



1.2 Ptolemy and positivity tests

Does it look familiar?

Ptolemy relation: $\overline{AC} \cdot \overline{BD} = \overline{AB} \cdot \overline{CD} + \overline{BC} \cdot \overline{AD}$



Consequence:

$$\left\{ \begin{array}{c} \text{efficient} \\ \text{positivity test} \end{array} \right\} \xleftrightarrow{1-1} \left\{ \begin{array}{c} \text{triangulation} \\ \text{of the } n\text{-gon} \end{array} \right\}.$$

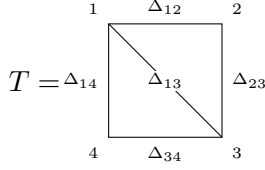
$\Rightarrow$ one triangulation contains the information of *all* positivity tests using *flips*.

1.3 The cluster algebra $\mathcal{A}_T$

$\rightsquigarrow$ We use this information to define a *cluster algebra* $\mathcal{A}_T$ associated to a triangulation T .

Definition 2. $\mathcal{A}_T$ is the subalgebra of $\mathbb{Q}(\Delta_{13}, \Delta_{12}, \Delta_{14}, \Delta_{23}, \Delta_{34})$ generated by all minors obtained from T using *flips*.

$$\mathcal{A}_T = \left\langle \Delta_{13}, \Delta_{12}, \Delta_{14}, \Delta_{23}, \Delta_{34}, \frac{\Delta_{12}\Delta_{34} + \Delta_{14}\Delta_{23}}{\Delta_{13}} = \Delta_{24} \right\rangle$$

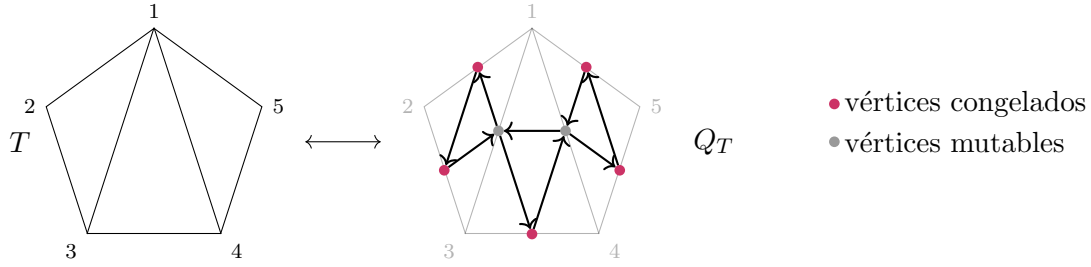


Note: $\mathcal{A}_T$ is the coordinate ring of the Grassmannian $\text{Gr}_2(4)$.

$$\text{Gr}_k(n) := \{V \subset \mathbb{K}^n : \dim_{\mathbb{K}}(V) = k\}.$$

Theorem 1 (Scott 2006). *The coordinate ring of the Grassmannian $\text{Gr}_k(n)$ in its Plücker embedding is a cluster algebra.*

1.4 Quiver and Quiver Mutation



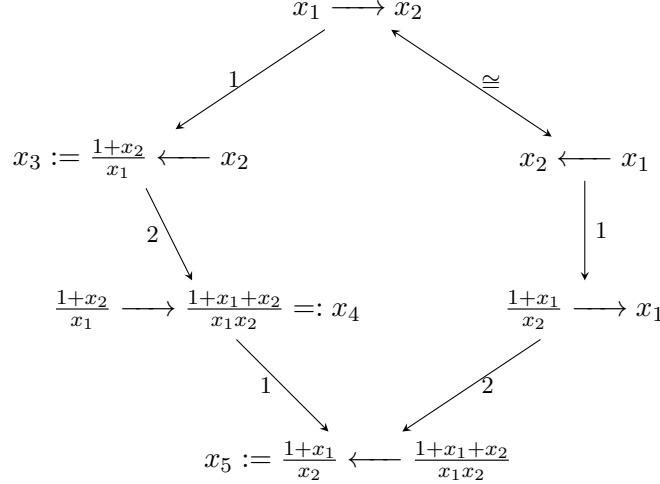
Definition 3 (Quiver mutation). *Let Q be a quiver without 1- and 2-cycles. The mutation $\mu_k(Q)$ of Q at mutable vertex k is obtained by: $\forall i \rightarrow k \rightarrow j$ add $i \rightarrow j$, reverse every arrow at k , and remove 2-cycles.*

$$\Delta_{ik}\Delta_{jl} = \Delta_{ij}\Delta_{kl} + \Delta_{il}\Delta_{jk} \quad \longleftrightarrow \quad x'_k x_k = \prod_{i \rightarrow k \in Q} x_i + \prod_{k \rightarrow j \in Q} x_j.$$

Definition 4 (Seed mutation). *Given a seed $s = (\mathbf{x}, Q)$ and a mutable vertex k of Q , the mutation in direction k is $\mu_k(s) = (\mu_k(\mathbf{x}), \mu_k(Q))$, where $\mu_k(\mathbf{x}) = \mathbf{x} \setminus \{x_k\} \cup \{x'_k\}$.*

1.5 Example: Seed Mutation

Recall the relation: $x'_k x_k = \prod_{i \rightarrow k \in Q} x_i + \prod_{k \rightarrow j \in Q} x_j$.



We define the *cluster algebra*:

$$\mathcal{A}_Q := \left\langle x_1, x_2, \frac{1+x_2}{x_1}, \frac{1+x_1+x_2}{x_1x_2}, \frac{1+x_1}{x_2} \right\rangle \subset \mathbb{Q}(x_1, x_2)$$

1.6 Fundamental Theorems

Theorem 2 (Laurent phenomenon (Fomin–Zelevinsky 2001)). *All cluster variables are Laurent polynomials in the cluster variables of the initial seed with integer coefficients. More precisely, they are contained in*

$$\mathbb{Z}[x_1^\pm, \dots, x_n^\pm, x_{n+1}, \dots, x_{n+m}].$$

Theorem 3 (Positivity Conjecture (Fomin–Zelevinsky 2001)). *All cluster variables are contained in*

$$\mathbb{N}[x_1^\pm, \dots, x_n^\pm, x_{n+1}, \dots, x_{n+m}].$$

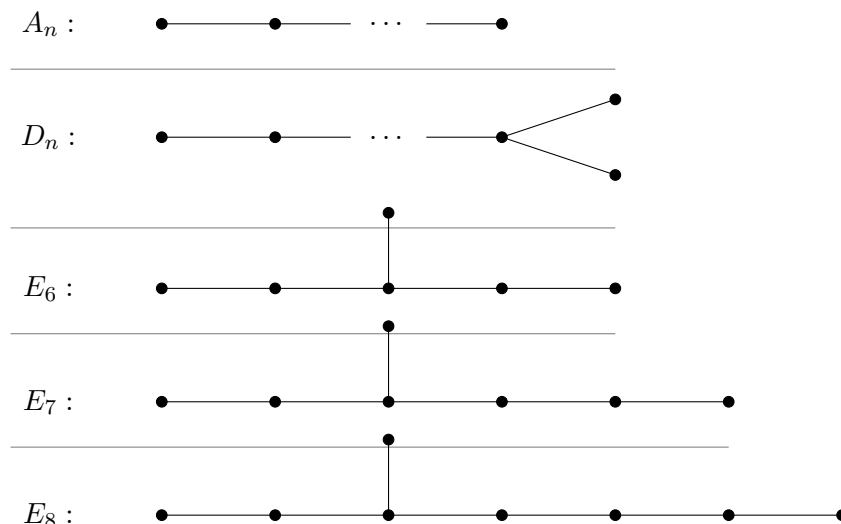
Theorem 4 (Positivity Theorem (Gross–Hacking–Keel–Kontsevich 2014)). *The positivity conjecture is true.*

Remark 1. *Gross–Hacking–Keel–Kontsevich view cluster algebras as rings of functions on log Calabi–Yau varieties (cluster varieties) and use tools from birational geometry and mirror symmetry.*

1.7 Cluster Algebras and Scattering Amplitudes

1.7.1 Finite type classification

Theorem 5 (Fomin–Zelevinsky 2003). *The cluster algebra $\mathcal{A}_Q$ is of finite type (the set of seeds is finite) if and only if (the mutable part of) Q is mutation equivalent to an orientation of a Dynkin diagram of type ADE:*



Remark 2. *The theorem above establishes a profound connection between the theory of cluster algebras and classical Lie theory. The mutation equivalence of the quiver Q corresponds exactly to the reflection groups associated with the simply-laced Dynkin diagrams.*

1.8 Examples of cluster structures

1. Markov triples

- are triples of integers (a, b, c) satisfying $a^2 + b^2 + c^2 = 3abc$,

- let $Q = \begin{matrix} & 2 \\ \nearrow & & \searrow \\ 1 & \xleftrightarrow{\quad} & 3 \end{matrix}$ and $(x_1, x_2, x_3) = (1, 1, 1)$,
- if $(\mathbf{x}', Q) \sim (\mathbf{x}, Q)$ then $\mathbf{x}' = (x'_1, x'_2, x'_3)$ is a Markov triple.

2. The Jones polynomial:

$$\begin{array}{ccccccc} \text{knot or link} & & \text{continued} & & \text{snake} & & \text{cluster} \\ \text{of 2 bridges } L & \rightsquigarrow & \text{fraction} & \rightsquigarrow & \text{graph} & \rightsquigarrow & \text{variable } x_L \end{array}$$

[Lee-Schiffler] The Jones polynomial of L (up to normalization by the leading term) is equal to the Laurent polynomial of x_L (up to a specialization of coefficients).

3. Scattering amplitudes...

1.9 Scattering amplitudes

In quantum field theory, physicists compute probabilities of particle interactions via scattering amplitudes.

The *scattering amplitude* is a function on the kinematic space that models the particle configuration, traditionally computed as an infinite sum of Feynman integrals (very hard).

Novel approaches study scattering amplitudes in complex coordinate systems (twistor theory), expressing them as iterated integrals and analysing:

- their singularity structure, encoded in their *symbol*
- recursive structures of amplitudes
- geometric interpretations, e.g. as volume of the *amplituhedron*

1.10 From particle configurations to algebraic varieties

Minkowski space $\mathbb{R}^4$ with inner product

$$p \cdot p' = p_0 p'_0 - (p_1 p'_1 + p_2 p'_2 + p_3 p'_3).$$

To $p \in \mathbb{R}^4$ we associate its *helicity spinor*

$$\sigma : p \mapsto \begin{pmatrix} p_0 + p_3 & p_1 - ip_2 \\ p_1 + ip_2 & p_0 - p_3 \end{pmatrix} \in \mathbb{C}^{2 \times 2}$$

with $\lambda = \frac{1}{\sqrt{p_0 + p_3}} \begin{pmatrix} p_0 + p_3 \\ p_1 + ip_2 \end{pmatrix}$, $\tilde{\lambda} = \frac{1}{\sqrt{p_0 + p_3}} \begin{pmatrix} p_0 + p_3 \\ p_1 - ip_2 \end{pmatrix} \in \mathbb{C}^2$ we have

$$\sigma(p) = \lambda \tilde{\lambda}^T.$$

Remark 3. The mapping σ represents the standard isomorphism between Minkowski spacetime $\mathbb{R}^{1,3}$ and the space of 2×2 Hermitian matrices. This forms the foundational link between physical momentum spaces and the algebraic spinor formalisms.

The mass m of p satisfies $p^2 := p \cdot p = m^2$. If p is *lightlike* $m = 0$, we have

$$\det(\sigma(p)) = p_0^2 - (p_1^2 + p_2^2 + p_3^2) = p^2 = 0.$$

Let $\{p_1, \dots, p_n\} \subset \mathbb{R}^4$ be a configuration of n lightlike particles satisfying *momentum conservation* $\sum_{i=1}^n p_i = 0 \implies$ helicity spinors are n pairs $(\lambda_i, \tilde{\lambda}_i) \in \mathbb{C}^{2 \times 2}$, defining

$$\Lambda = [\lambda_1, \dots, \lambda_n], \quad \tilde{\Lambda} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n] \in \text{Gr}_{2;n}$$

Exercise: Momentum conservation translates to $\Lambda \tilde{\Lambda}^T = 0$.

Definition 5. The spinor helicity variety is defined as

$$\mathcal{SH}_n := \{(\Lambda, \tilde{\Lambda}) \in \text{Gr}_{2;n} \times \text{Gr}_{2;n} : \Lambda \tilde{\Lambda}^T = 0\}.$$

$\mathcal{SH}_n$ is parametrized by 2×2 minors with $1 \leq i < j \leq n$

$$P_{ij} := \det(\lambda_i \lambda_j) \quad \text{and} \quad \tilde{P}_{ij} := \det(\tilde{\lambda}_i \tilde{\lambda}_j)$$

satisfying *determinantal identities* and *momentum conservation*

$$\begin{aligned} 0 &= P_{ij} P_{kl} - P_{ik} P_{jl} + P_{il} P_{jk} = \tilde{P}_{ij} \tilde{P}_{kl} - \tilde{P}_{ik} \tilde{P}_{jl} + \tilde{P}_{il} \tilde{P}_{jk} \\ 0 &= \sum_{s=1}^n P_{is} \tilde{P}_{sj} \quad (\Leftrightarrow \Lambda \tilde{\Lambda}^T = 0) \end{aligned}$$

Remark 4. The determinantal identities $P_{ij} P_{kl} - P_{ik} P_{jl} + P_{il} P_{jk} = 0$ are exactly the classical Plücker relations governing the projective embedding of the Grassmannian.

1.11 Some results

Theorem 6 (B.–Jianrong Li). *The homogeneous coordinate ring of the spinor helicity variety is a cluster algebra that embeds into a Grassmannian cluster algebra.*

Theorem 7 (B.–Drummond-Glew, B.–Drummond-Glew–Gürdoğan-Wright, Pokraka–Spradlin–Volovich–Weng). *For five or less particles the scattering amplitude is uniquely determined by the cluster algebra. For six particles, the symbol is partially given by cluster variables.*

Theorem 8 (Galashin). *The (real) spinor helicity variety contains a positive subset — the (momentum) amplituhedron with canonical volume form whose integral is the scattering amplitude.*

1.12 Symbol of a polylogarithm

Example 2. $Li_k(z) := \sum_{j=1}^{\infty} \frac{z^j}{j^k}$ is a polylogarithm satisfying

$$Li_k(z) = \int_0^z Li_{k-1}(t) d \log t, \quad Li_1(z) = -\log(1-z)$$

The symbol can be computed recursively: $\mathcal{S}(Li_1(z)) = -(1-z)$ and

$$\mathcal{S}(Li_k(z)) = -(1-z) \otimes (z)^{\otimes k-1}.$$

Remark 5. *The symbol $\mathcal{S}$ maps iterated integrals modulo functions of lower transcendentality into a tensor algebra. It serves as an algebraic invariant simplifying the derivation of highly nontrivial polylogarithmic identities essential in quantum field theories.*